

2023 Ch H1 Q24

Section: Chemistry in Society

Topic: The Mole Concept

Question Summary

The number of moles of positive ions in 0.25 moles of aluminium sulfate, $\text{Al}_2(\text{SO}_4)_3$, is required.

Worked Solution

Step 1: Write formula of aluminium sulfate: $\text{Al}_2(\text{SO}_4)_3$.

Step 2: Each formula unit contains 2 Al^{3+} ions and 3 SO_4^{2-} ions.

Step 3: For 1 mole of $\text{Al}_2(\text{SO}_4)_3$, there are 2 moles of Al^{3+} ions.

Step 4: For 0.25 moles of $\text{Al}_2(\text{SO}_4)_3$, the moles of Al^{3+} ions = $0.25 \times 2 = 0.50$ mol.

Final Answer

A — 0.5 mol of Al^{3+} ions

Revision Tips

- Identify the number of each ion in a formula unit.
- Multiply the number of ions per unit by the moles of compound.
- Always double-check subscripts in ionic compounds.